

Telecom Customer Churn Prediction - Project Summary

Project Objective

The objective of this project is to predict telecom customer churn and identify customers who are likely to discontinue telecom services. The model helps businesses improve customer retention by proactively targeting high-risk customers.

Dataset & Preprocessing

- Loaded and inspected telecom customer data.
- Performed data cleaning and handled missing values.
- Conducted feature engineering to improve predictive performance.
- Encoded categorical variables using suitable encoding techniques.
- Standardized numerical features where required.

Exploratory Data Analysis (EDA)

The project analyzed customer demographics and service usage patterns including age, salary, call activity, SMS usage, and data consumption. EDA was used to identify trends, class imbalance, feature relationships, and churn-related behavior.

Machine Learning Pipeline

- Data preprocessing and feature preparation.
- Handling class imbalance using SMOTE.
- Model training and evaluation using multiple machine learning techniques.
- Performance measurement using Accuracy, ROC-AUC Score, and Confusion Matrix.

Algorithms Identified in the Notebook

- Logistic Regression
- Random Forest Classifier
- Additional preprocessing and evaluation utilities

Business Impact

The final solution helps telecom companies identify customers at risk of churn, enabling targeted retention campaigns, reduced customer loss, improved customer satisfaction, and better revenue protection.

Key Skills Demonstrated

Python, Pandas, NumPy, Data Cleaning, Feature Engineering, Data Visualization, Exploratory Data Analysis, Machine Learning, Classification Modeling, Model Evaluation, and Business Analytics.